CCCA Meta-Atlas · Cell-State Analysis

Caron et al. 2020 — Pediatric Acute Lymphoblastic Leukemia (10x, 8 malignant samples, 25,788 cells)

GO / NO-GO GATES

Gate 1 · 01→02 · GO

Malignant manifold is sample-mixed AND continuous. 8 patients across 3 genetic ALL groups (ETV6-RUNX1 Pre-B, high-hyperdiploid B-ALL, Pre-T) merge on ONE connected component after Harmony (θ=10) on `sample`. iLISI 1.00→4.71 of 8. Leiden split is proliferation (cycling 0.87 vs 0.06), not B-vs-T lineage; 0 islands removed.

Gate 2 · 03→04 · GO (non-branching)

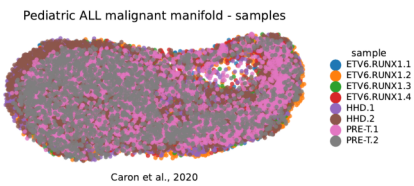
Embryonic pattern present as organized cross-repression, NOT a convergence hub. Gast & pSom sit at opposite poles (smoothed r=−0.45; joint-high enrichment 0.81<1); Crest is an inverse/terminal gradient (pSom–Crest r=−0.72). Root Gast-high (ρ=−0.72 vs pseudotime) → terminus Crest-high (ρ=+0.33). No bifurcation.

Gate 3 · 04→05 · NA

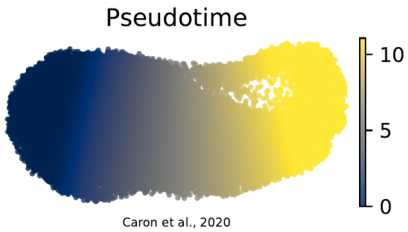
Not evaluated — non-branching topology (principal curve, 2 tips, 0 forks, fold-free). No tree node-clusters, so no mutual Gast↔pSom cross-repression test applies. Terminates at the expression step.

STEP FIGURES

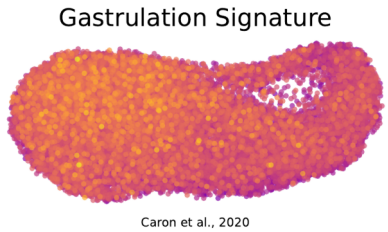
01 · Samples (integration)



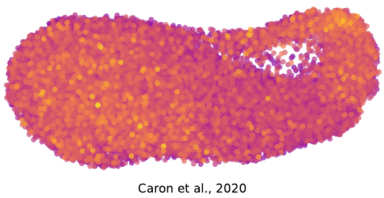
02 · Pseudotime



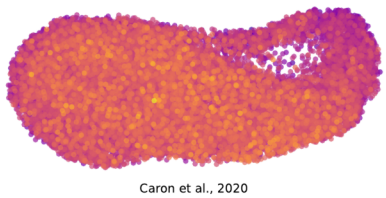
03 · Gastrulation (raw)



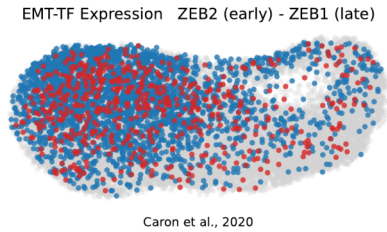
03 · Somitogenesis (raw)



03 · Neural Crest (raw)



04 · EMT-TF expression



PARAMETER CHOICES & RATIONALE

- Min-cell threshold: floor ≥100 (gap-finder found no dominant spillover cliff; all 8 malignant samples ≥1585) → 8/8 kept, 25,788 cells.
- Integration: Harmony on `sample`, θ=10 (same-sample neighbour frac 0.999→0.272; strong batch structure from 3 genetic ALL groups).
- Embedding: PCA 50→20 PCs on X_pca_harmony, neighbours k=30, UMAP spread=1.3, seed=1. Island removal: 6× median-kNN radius graph (0 removed).
- Geometry: principal CURVE (scFates EPG, Nodes=20, λ=0.3, μ=100) — PPT tree over-forks a dense blob (11–13 spurious forks); curve fold-free.
- Root: curve tip 0 (Gast/pSom-high, Crest-low) — developmental origin, never Crest (late) or a cycling state.
- Modules score_genes with union control pool (Gast 66/81, pSom 77/98, Crest 93/104 genes); panels show RAW per-cell scores.
- EMT-TFs (detectable >3%): ZEB2 (40.6%), ZEB1 (8.9%), SNAI1 (6.8%); FOXC/PRRX/SNAI2/TWIST absent. Overlay pair ZEB2↔ZEB1.

VERDICT

Pattern reproduces as the NON-BRANCHING case: a single embryonic axis with organized Gast↔pSom cross-repression (r=−0.45) and an inverse terminal neural-crest gradient — a strict linear subset of the Puram hub-and-spoke, with no bifurcation. Root Gast/pSom-high → terminus Crest-high. Gate 1 GO / Gate 2 GO / Gate 3 NA.